

Original CCI-Class Controller — Project Roadmap

CEI 0-16 Annex O/T/M compliant Central Plant Controller

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Abstract

Target: CEI 0-16 Annex O/T/M compliant Central Plant Controller, flexible 100 kW–6 MW MT plant range.

Scope: Original hardware/firmware design (not a clone of any existing commercial product).

Prototype budget: ~€1,000 (single unit, BOM + fab + assembly).

Timeline stance: Balanced — solid working prototype in ~10–12 weeks; grid/cyber certification is a separate, longer track.

Existing certification: organization already holds **ATEX** (EN/IEC 60079) — reuse QMS, Notified Body relationship, and technical-file practice for the CEI 0-16 track below.

Alternate track: see companion brief **CCLI-CLONE-RE-001** (CCI_Clone_RE_Option.pdf) for the clone / reverse-engineering option, risks, and decision guide.

Already in place: your company is **ATEX-certified** (EN/IEC 60079 product family experience). That is a real head start — technical files, NB engagement, and production QA patterns do not start from zero. Standard MT CCI cabinet installations are usually *non-hazardous area*; confirm per project whether Ex marking is required on the CCI itself.

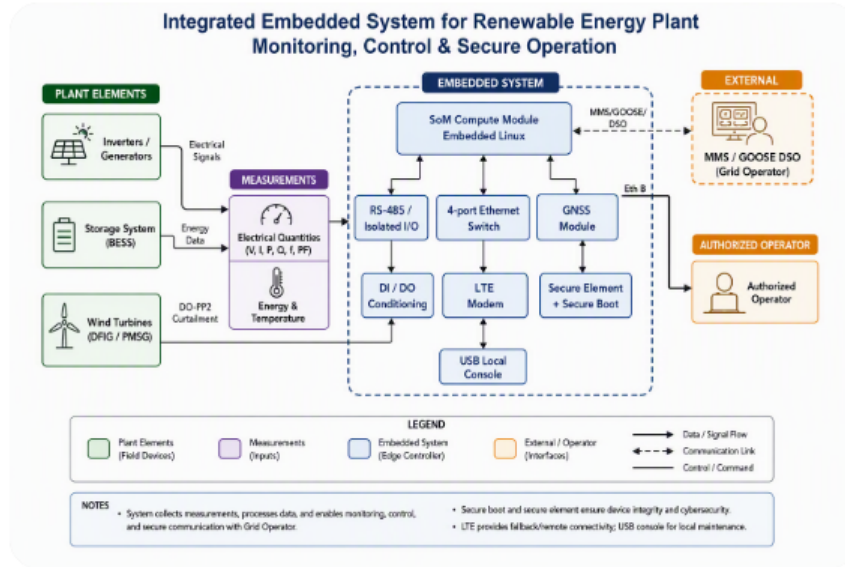
Set expectations up front: the hardware prototype is the *cheap and fast* part of this project. Certification still required for this product class (IEC 62443-4-1/4-2, IEC 61850 conformance, IEC 62351, CEI 0-16 demonstration) — realistically €50,000–120,000+ and 9–18 months for the *remaining* grid/cyber scope. ATEX is already held and is not budgeted again here.

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1 System architecture (original design)

This block diagram reflects the *generic, publicly-documented functional interfaces* that CEI 0-16 Annexes O/T require of any CCI (dual DSO/Operator Ethernet separation, plant-side ports, serial link to the energy analyzer, time sync, DI/DO for status/command) — not any specific vendor’s schematic or component selection. Figure 1 shows the visual companion to this section.



I've redesigned it into a cleaner presentation-ready architecture chart with:

Figure 1: CCI system architecture — integrated embedded edge controller for renewable plant monitoring, control, and secure DSO/operator communication (CEI 0-16 Annexes O/T functional view)

1.1 Core functional blocks

2 Compute module: SoM + custom carrier

Decision (from our discussion): off-the-shelf industrial SoM, original carrier board design.

Candidate SoM families to evaluate (pick based on Linux BSP maturity, industrial temp range, and 10-year longevity commitment — you need this for a utility-facing product):

- Toradex Verdin iMX8M Mini/Plus
- Variscite DART-MX8M-Mini / DART-MX8M-Plus
- Compulab CL-SOM-iMX8
- NXP i.MX8M Mini EVK-class reference designs

All of these ship a maintained Yocto BSP, handle DDR/power-sequencing/thermal internally, and let your carrier board stay a straightforward 2–4 layer design (Ethernet PHYs, RS-485 transceivers, GNSS/LTE module sockets, DI/DO conditioning, power).

Table 1: Core functional blocks and design-track assignment

Block	Function	Notes / track
Design approach	Option A (default): original SoM + carrier derived from public CEI 0-16 Annex O/T interfaces	Primary track for this roadmap. Independent schematic, layout, and firmware boundary.
Design approach	Option B: clone via reverse engineering of a commercial CCI (AiLux-class reference)	Separate brief CCLI-CLONE-RE-001 (CCI_Clone_RE_Option.pdf). Higher IP/legal and certification risk; use only as a gated research spike, not the default product path.
Compute module (SoM)	Runs embedded Linux, IEC 61850 server, Modbus master, PF2 control logic	Option A: Toradex/Variscite/Compulab-class SoM. See §2. Option B: match reference compute block only if legally cleared.
Ethernet interfaces (≥ 4 -port managed switch or SoM+switch IC)	Eth_A $\rightarrow$ DSO (isolated, FO-ready via external media converter); Eth_B $\rightarrow$ authorized Operator; $2\times$ plant-facing ports	Functional requirement: physical/electrical separation between DSO and plant networks — minimum two isolated PHYs/VLANs. Do not copy vendor PHY placement verbatim on Option B.
RS-485/RS-232 serial ($\times 2$, isolated)	Modbus RTU master to energy analyzer (Ax M22/MT1-class); IEC 60870-5-101 optional	Opto-isolated transceivers, switchable termination. Map registers from analyzer datasheet, not from copied reference firmware.
GNSS module	Time sync for timestamped measurements and events	Standard u-blox-class module with passive/active antenna. Commodity block on both tracks.
Cellular modem (LTE Cat-1/Cat-4)	Remote configuration, monitoring, and backup communications	Optional on rev A if DSO link is wired. Modular socket; omit from first prototype if not needed.
Digital I/O	DI: polarized/optoisolated 10–120 Vdc status inputs; DO: dry contact for command/curtailment relay	Critical PF2 actuation path. Option A: design from CEI limits. Option B: observe reference thresholds only; re-implement conditioning network.
Secure element + secure boot	Root of trust, key storage, and signed firmware	Option A: ATECC608B/STSAFE-A110-class parts with your own key hierarchy. Option B: do not extract or reuse reference keys/boot chain.
Local console	USB service port for commissioning and diagnostics	Outside certified control path on both tracks; web/Python tooling acceptable here.
Power input	12–24 Vdc wide-range isolated DC-DC	Typical Box CCI UPS/battery context. Size from SoM + field I/O budget; standard off-the-shelf power module.

3 Prototype BOM target (single unit, ~€1,000)

Item	Est. cost (EUR)	Notes
Industrial SoM (qty 1)	€ 60–120	Toradex/Variscite-class, i.MX8M Mini tier
Carrier PCB fab (4-layer, small batch 5–10 pcs)	€ 50–150	JLCPCB/PCBWay prototype tier
Assembly (self or light-touch SMT service)	€ 100–250	BGA stays on the SoM module — carrier assembly is hand/reflow-feasible
Ethernet PHYs/magnetics/RJ45 (×4)	€ 15–25	u-blox-class Quectel-class; optional on rev A e.g. Microchip ATECC608B / STSAFE-A110 class part
RS-485/RS-232 isolated transceivers (×2)	€ 10–15	
GNSS module + antenna	€ 25–40	
LTE module + antenna + SIM holder	€ 25–40	
Secure element	€ 2–5	
DI/DO conditioning (opto-isolators, relays)	€ 15–25	
Power stage (wide-range isolated DC-DC)	€ 15–25	
DIN rail hardware, connectors, terminal blocks, enclosure	€ 40–60	
Passives, stencil, shipping, contingency	€ 80–150	
Total (single prototype)	~€ 440–900	Leaves headroom within €1,000 for one respin's worth of parts

This is a *component-cost* estimate for one unit at prototype quantities — not a manufactured unit cost at volume, which would drop significantly (likely €150–300/unit at 100+ qty once NRE is amortized).

4 Path to Gerbers

5 Firmware architecture & language choice

- **OS/BSP:** Yocto-based embedded Linux (matches SoM vendor's supported BSP) — gives you package management, OTA update tooling, and a known-good foundation auditors are familiar with.
- **Core control/protocol path** (IEC 61850 server, Modbus master, PF2 curtailment logic, GOOSE handling): C/C++. This is the code that will eventually face conformance and security audits — auditors and test labs are far more comfortable reviewing statically-typed, analyzable C/C++ than dynamic scripting languages, and real-time GOOSE timing requirements favor it anyway.
- **Config/commissioning tooling, local web console, diagnostics:** Python or a lightweight web stack is fine here — this code is outside the certified control-path boundary, so language

Table 2: Carrier board workflow — Altium Designer to prototype fab

#	Step	Activity (Altium Designer)
1	Architecture freeze	Lock interfaces, connector count, and power budget before opening the project — changes after layout start cost a full re-pass.
2	Schematic capture	Capture in Altium Designer from datasheets and your interface decisions. Link SoM/PHY/transceiver symbols to verified footprints early.
3	Design review	Checklist pass: power sequencing vs SoM requirements; ESD/surge on every field I/O (RS-485, DI/DO, Ethernet); isolation barriers where required.
4	PCB layout	Place SoM footprint first (follow vendor carrier guide), then Ethernet PHY/magnetics, then RS-485, power, and RF/LTE. Keep analog/field wiring away from switching nodes.
5	DFM check	Run Altium DRC, then the target fab’s DFM portal (JLCPCB/PCBWay) against fab stackup and drill rules before order.
6	Fab outputs	Generate via Altium Output Job : Gerber RS-274X, NC drill, IPC-2581 or ODB++ if requested, assembly pick-and-place, and BOM (ActiveBOM or structured export).
7	Fab + assembly	1–3 week turnaround typical at prototype scale; hand or SMT assembly on carrier only (BGA stays on SoM).
8	Bring-up	Power-on test, boot SoM, verify each interface individually (Ethernet ×4, RS-485 ×2, GNSS, LTE, DI/DO) before firmware integration.

choice is a productivity call, not a compliance one. Keep the boundary between “certified path” and “tooling” architecturally clean from day one — it makes scoping the IEC 62443-4-2 assessment much cheaper later.

- **IEC 61850 stack:** start on **libiec61850** (MZ Automation, MPL license) for prototype/demo. Budget to evaluate a commercial stack (SISCO, Triangle MicroWorks) before booking paid conformance testing — see §7.
- **Security baseline to build in from day one** (cheap now, expensive to retrofit later): secure boot chain, signed firmware images, TLS/DTLS for IEC 62351, key storage in the secure element, basic RBAC on the local console, audit logging. IEC 62443-4-1 specifically audits *how* you built and maintain the product (your SDLC), not just what features it has — so start documenting your development process now, not after the fact.

6 Timeline — Phase 1: Hardware + firmware prototype (~10–12 weeks)

Weeks	Milestone
1–2	Architecture freeze, SoM selection finalized, schematic capture starts
2–4	Schematic complete + design review, long-lead parts ordered (SoM, LTE/GNSS modules)
4–6	PCB layout, DFM check, Gerbers to fab
6–8	Fab + assembly turnaround
8–10	Hardware bring-up: power, boot, verify each interface (Ethernet ×4, RS-485 ×2, GNSS, LTE, DI/DO)
10–12	Firmware integration: libiec61850 server running, Modbus master polling a real analyzer, DO-driven curtailment stub, local console reachable

Exit criteria for this phase: a DIN-rail prototype that boots reliably, exposes an IEC 61850 MMS server over Eth_A, polls a Modbus energy analyzer over RS-485, timestamps data via GNSS, and can toggle a DO relay from a simulated DSO command. This is a credible demo/investor/pilot-partner artifact — it is *not* yet field-deployable at a real MT POC.

7 Phase 2: Certification path (separate track, 9–18 months)

Scope note: ATEX (EN/IEC 60079) is **already certified** for the organization — treat it as established capability, not a new line item. Phase 2 below covers the *remaining* CEI 0-16 / grid-interface / cybersecurity workstreams for this CCI product.

Table 3: Certification workstreams — remaining scope (ATEX already held)

Workstream	What it involves	Rough cost	Duration
ATEX (EN/IEC 60079)	Already held — existing org certification, NB relationship, and hazardous-area design process	Sunk / reuse	—
IEC 62443-4-1 (Secure SDLC)	Process audit of secure development lifecycle. Partial overlap with ATEX QMS/documentation habits, but cyber-specific artifacts still required	€10,000–30,000	2–5 months
IEC 62443-4-2 (device security)	Lab test vs SL1 (CEI 0-16 FAQ minimum), target SL2 capability	€15,000–40,000	2–4 months
IEC 61850 conformance	Accredited lab (DNV/KEMA, OMI-CRON, Kalkitech-class) validates MMS/GOOSE/reporting	€10,000–25,000	1–3 months
IEC 62351 protocol testing	Cybersecurity protocol testing; scarce lab capacity — schedule slack	Overlaps 4-2	3–6 months
CEI 0-16 Annex O/T + DSO demo	Functional observability / PF2 exchange with DSO or recognized test environment	Integrator time	1–3 months

All-in realistic budget for first CEI-class CCI certification: roughly €45,000–110,000, spread over 9–18 months (ATEX excluded — already held). This is the real reason the commercial Italian CCIs (AiLux, Selta, Energy Team, Higeo) price the way they do — you’re not paying for a €400 BOM, you’re amortizing exactly this certification investment across thousands of installed units. Your path to being genuinely cheaper is either (a) volume — this cost amortizes the same way once you’re past unit 50–100 — or (b) a leaner process by scoping the 62443-4-2 boundary tightly around just the certified control path (see the firmware architecture note in §5), which measurably shrinks lab scope and cost.

8 Team/skills needed

- Embedded hardware engineer (schematic + layout, comfortable with SoM carrier design)
- Embedded Linux/firmware engineer (Yocto, C/C++, protocol stacks)
- Someone who can own the IEC 62443-4-1 process documentation — can leverage existing ATEX technical-file and NB workflow experience
- A relationship with an accredited test lab early — your ATEX Notified Body may also offer or partner on 62443/61850 scoping

9 Immediate next steps

1. Confirm SoM family choice (Toradex vs Variscite vs Compulab) based on your BSP/tooling comfort and long-term sourcing preference.
2. Draft the interface list and pin budget (connector-by-connector) as a starting schematic outline — original to your design, not copied from any existing product.
3. Get a scoping quote from one IEC 62443/61850 test lab now, even before hardware exists — their capacity and requirements should shape your firmware architecture decisions, not the other way around.